Ojasvi Shrivastava

Data Science @UC Berkeley | Machine Learning · Generative AI · Data Science | Building AI-Powered Solutions

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Education

University of California, Berkeley Aug 2025 – May 2027
B.A. Data Science *Berkeley, CA*

Los Angeles Pierce College Jan 2023 – Jul 2025
A.S. Mathematics A.S. Physics A.S. Computer Science A.A. General Studies *Los Angeles, CA*

Certifications

Intermediate Machine Learning | *Kaggle* | XGBoost, cross-validation, data leakage prevention, feature encoding

Intro to Deep Learning | *Kaggle* | TensorFlow, Keras, neural network architecture, performance tuning

Artificial Intelligence Fundamentals | *IBM* | ML principles, AI ethics, applied AI systems

Technical Skills

Languages: Python, SQL, Java, C++, TypeScript, JavaScript

AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LLMs, RAG, NLP, Deep Learning, Agentic Workflows, Vector Databases, Feature Engineering, Cross-Validation, Monte Carlo Simulation

Data Science & Analytics: NumPy, Pandas, Matplotlib, Plotly.js, GeoPandas, Statistical Modeling, Stochastic Simulation, Optimization Models

Frameworks & Libraries: Next.js, React.js, FastAPI, Tailwind CSS, Leaflet.js, Vercel

Developer Tools: GitHub, VS Code, Jupyter Notebook, Overleaf

Experience

Private Mathematics Tutor Jan 2024 – Sep 2025

- Designed first principles problem solving frameworks for Calculus, Linear Algebra, and Statistics, enabling students to derive results from definitions rather than memorized formulas, achieving an average 60% exam score improvement across tutees.
- Diagnosed individual learning blockers (algebra fluency gaps, proof technique deficiencies, conceptual misalignments) and re-sequenced curricula accordingly, building a reusable library of targeted problem sets refined over 20+ sessions.

Projects

Scout | *Autonomous AI Market Intelligence Engine* | Python, LLMs, RAG, Agentic Workflows, NLP, Vector DBs, FastAPI

- Architected an autonomous AI market intelligence engine ingesting global signals research papers, patents, startup launches, VC funding, job postings, and government grants to surface high-value, unserved opportunities before they become obvious.
- Engineered an agentic pipeline using LLMs and RAG to auto-generate startup concepts, estimate TAM/SAM/SOM market sizing, perform competitive analysis, draft MVP specs, and produce investor pitch decks.
- Shifted paradigm from reactive Q&A to proactive opportunity discovery via multi-source data fusion across structured and unstructured corpora at scale.

GIStice League | *Machine Learning Team* | *UC Berkeley Open Project* | Python, PyTorch, scikit learn, GeoPandas, NumPy

- Contributed ML modeling to a UC Berkeley open project mapping grocery accessibility across 500+ census tracts in SF, Oakland, and Berkeley using PyTorch and scikit learn regression models.
- Engineered features cross referencing an Accessibility Index and Residual Gradient against 10+ demographic variables (racial distribution, median income, homeownership rates, population density) to identify food deserts.
- Built interactive Leaflet.js and Plotly.js visualizations enabling spatial equity analysis across Bay Area socioeconomic strata.

FinSight | *Full-Stack Wealth & Retirement Simulator* | Python, Next.js, TypeScript, React.js, Tailwind CSS, Monte Carlo, Vercel

- Developed a full-stack wealth and retirement simulator using Next.js and TypeScript, deploying stochastic Monte Carlo simulations to model portfolio outcomes across thousands of market scenarios.
- Implemented AI driven expense categorization and dynamic budget trend visualizations, enabling data driven retirement planning through interactive dashboards deployed on Vercel with a FastAPI backend.